

Learning When to Stop: Kelly-Optimal E-Value Portfolios for Early Multi-Stream Vaccine Safety Surveillance

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Abstract—Europe’s first major pauses of the Oxford–AstraZeneca COVID-19 vaccine came after roughly 5–5.5 million exposures had accrued in the EU/EEA. We ask whether a joint, anytime-valid monitor could have raised an earlier alarm. We model post-deployment safety surveillance as a portfolio of e-value streams, one for each adverse event of special interest, and learn how to allocate inferential capital across them while preserving validity under continuous monitoring. The objective is to minimize expected time to threshold crossing; for large thresholds this is equivalent to maximizing expected log growth, linking sequential safety monitoring to Kelly-optimal portfolio choice. In an AstraZeneca-calibrated retrospective exposure-to-alarm simulation based on 13 monitored outcomes, the learned portfolio crosses its statistical threshold after about 0.63 million vaccinations under immediate observation, and after about 2.6 million vaccinations after a conservative lag adjustment anchored to Guillain–Barré syndrome onset. This is roughly half of the 5–5.5 million exposures accrued before the first major European pause.

Index Terms—E-values, e-processes, optional stopping, pharmacovigilance, portfolio optimization, Kelly criterion, vaccine safety, AstraZeneca, sequential analysis.

I. INTRODUCTION

IN early 2021 the world was still deep in the acute phase of the COVID-19 pandemic. Vaccines had been developed at unprecedented speed, and mass deployment had become one of the central public-health tasks of the century. In Europe, the adenoviral-vector Oxford–AstraZeneca vaccine received conditional marketing authorization from the European Medicines Agency (EMA) on 29 January 2021, and use in the EU/EEA began in the first week of February [1]. Only weeks later, concern shifted from scarcity to surveillance. On 11 March 2021, the EMA reported that 30 thromboembolic events had been notified among close to 5 million people vaccinated in the EEA, while several countries had already begun pausing the vaccine [2]. By 15 March, Germany, France, Italy, Spain and others had joined the suspensions. The resulting sequence of pauses, reviews, and public debate made vivid a statistical fact that is easy to state but hard to operationalize: safety

monitoring is a sequential decision problem in which delay means further exposure.

The statistical difficulty was not merely that the suspected harms were rare. It was that evidence was dispersed across multiple adverse-event streams—neurological, thrombotic, hematological, and cardiac—with no guarantee that any single endpoint would carry the signal alone. The later multinational Global Vaccine Data Network (GVDN) analysis of 99 million vaccinated individuals confirmed exactly this structure. For first-dose ChAdOx1, some adverse events of special interest (AESI), including Guillain–Barré syndrome (GBS), cerebral venous sinus thrombosis (CVST), idiopathic thrombocytopenia and pulmonary embolism, were elevated relative to expected background rates, whereas many other monitored outcomes were not [3]. This is precisely the setting in which a surveillance system should not be forced to choose between watching one endpoint too narrowly and watching many endpoints too diffusely.

This paper develops an anytime-valid solution based on *e-values*. The main idea is simple. Each monitored harm generates a stream of valid evidence against its own null of no excess risk. Those evidence streams can then be combined sequentially as a *portfolio*. A predictable, data-dependent allocation of inferential capital across streams still produces a valid global e-process. Hence one may look continuously, stop whenever the accumulated evidence crosses a threshold, and still retain a finite-sample guarantee against false alarms.

The guarantee comes from a classical inequality due to Ville. If $(K_t)_{t \geq 0}$ is a non-negative supermartingale under the null with $\mathbb{E}_0[K_0] \leq 1$, then for every $\alpha \in (0, 1)$,

$$\mathbb{P}_0 \left(\sup_{t \geq 0} K_t \geq \frac{1}{\alpha} \right) \leq \alpha. \quad (1)$$

This is the mathematical reason that *peeking* is allowed: the probability of ever crossing the alarm boundary is controlled over the entire path, not merely at a single pre-fixed analysis time. E-values are designed exactly for this purpose [5], [6], [7]. Under the null, an e-value has expectation at most one. By multiplying one-step e-factors over time, one obtains an e-process, which is automatically a non-negative supermartingale. The resulting evidence

can therefore be monitored continuously without inflation from optional stopping.

The methodological contribution of the paper is to turn that validity principle into a *portfolio optimization* problem. Let K_t denote the current inferential capital of a global e-process and define the alarm time

$$\tau_\alpha := \inf\{t \geq 1 : K_t \geq 1/\alpha\}. \quad (2)$$

The operational goal is not “maximize expected return” or “minimize variance” in the abstract, but rather

$$\min_{\pi \in \Pi} \mathbb{E}_1[\tau_\alpha(\pi)], \quad (3)$$

where π is a portfolio rule, Π is the class of admissible predictable allocations, and $\mathbb{P}_1$ denotes a working alternative describing the world in which some harms are genuinely elevated. For large thresholds, minimizing expected time to hit $1/\alpha$ is approximately equivalent to maximizing expected log growth of the e-process. Thus the efficiency layer of the problem is Kelly-optimal: one should allocate inferential capital to maximize the growth rate of evidence [8], [9], [10]. The setting differs from gambling only in interpretation: the “wealth” being reallocated is not money, but valid evidence.

The substantive contribution is a retrospective case study. We calibrate a stylized patient-by-patient simulation to the GVDN ChAdOx1 study using 13 AESI streams. The study is best understood as an *exposure-to-alarm* analysis. That is, we ask how many vaccinations would have accrued before an anytime-valid monitor crossed its alarm threshold in a world whose event rates are calibrated to the observed-vs-expected study. We do *not* attempt a full calendar-time replay of reporting, adjudication, and regulatory deliberation. In our view this is a strength, not a weakness. For rare events, assuming approximate independence across individuals and streams is a realistic first-order approximation, and the estimand—vaccinations accrued before alarm—is directly relevant to public health.

The headline result is straightforward. In the immediate-observation simulation, the learned e-value portfolio crosses at about 0.63 million vaccinated individuals. To compare more fairly with the historical pauses, we then add a conservative lag adjustment anchored to GBS onset, the slowest of the elevated serious streams we monitor. This produces an operational alarm at about 2.6 million vaccinations. The real-world European pauses occurred after roughly 5–5.5 million exposures had accrued. The comparison is not a claim that regulators literally should have acted on a particular day; it is a claim that *the statistical architecture of surveillance matters*. Learning to combine weak but coherent evidence across many harms could, in principle, have made the signal legible materially earlier.

II. E-VALUES FOR SEQUENTIAL MULTI-STREAM SURVEILLANCE

A. What is an e-value in this setting?

Suppose that after each vaccinated individual we update m monitored safety streams. For stream $j \in \{1, \dots, m\}$,

let $Y_{t,j} \in \{0, 1\}$ indicate whether the AESI occurs in the chosen risk window for the t th exposed individual. The null hypothesis for stream j is that its event probability equals a known or estimated background rate p_{0j} . A one-step e-factor for stream j is any non-negative random variable $E_{t,j}$ satisfying

$$\mathbb{E}_0[E_{t,j} \mid \mathcal{F}_{t-1}] \leq 1, \quad (4)$$

where $(\mathcal{F}_t)$ is the natural information filtration. In the Bernoulli setting a convenient choice is a likelihood-ratio e-factor:

$$E_{t,j} = \left(\frac{q_{t-1,j}}{p_{0j}} \right)^{Y_{t,j}} \left(\frac{1 - q_{t-1,j}}{1 - p_{0j}} \right)^{1 - Y_{t,j}}, \quad (5)$$

with a predictable design probability $q_{t-1,j} \geq p_{0j}$. Under the null, the conditional expectation equals one exactly. Intuitively, $E_{t,j}$ rewards observations that are more likely under an elevated-risk working model than under the background-rate null.

A full e-process for stream j is then obtained multiplicatively:

$$K_{t,j} = \prod_{s=1}^t E_{s,j}, \quad K_{0,j} = 1. \quad (6)$$

Because $\mathbb{E}_0[E_{t,j} \mid \mathcal{F}_{t-1}] \leq 1$, the process $(K_{t,j})$ is a non-negative supermartingale under the null. That is the essential link between the stream-specific evidence and the global optional-stopping guarantee.

Definition 1. A process $(X_t)_{t \geq 0}$ adapted to $(\mathcal{F}_t)$ is a non-negative supermartingale under $\mathbb{P}_0$ if $X_t \geq 0$ almost surely and

$$\mathbb{E}_0[X_t \mid \mathcal{F}_{t-1}] \leq X_{t-1} \quad \text{for all } t \geq 1. \quad (7)$$

The logic of e-values is especially natural in pharmacovigilance. Safety monitoring is inherently sequential: data arrive as vaccination proceeds. It is also inherently adaptive: committees look when they choose, analysts update models when new data arrive, and the endpoints that appear most informative may change over time. A test whose interpretation breaks under repeated looks is therefore mismatched to the problem. E-values avoid that mismatch because the whole path is already calibrated through (1).

B. How a safety stream produces evidence

The e-value construction is especially transparent in the present setting. Each monitored stream starts from a background-rate null p_{0j} , typically derived from external historical data or an observed-versus-expected design. A working alternative $q_{t-1,j} \geq p_{0j}$ represents the possibility that the stream is elevated. The one-step factor in (5) then behaves intuitively. If an AESI occurs, the factor jumps by roughly $q_{t-1,j}/p_{0j}$, which can be substantial for rare but enriched harms. If the event does not occur, the factor is slightly below one, reflecting the fact that the alternative allocated some probability mass to an event that did not happen.

For example, the GVDN calibration gives a null rate of about 6.33 GBS cases per million first-dose ChAdOx1

recipients and an observed rate of about 15.83 per million. Under a design aligned with that observed rate, a GBS event contributes a factor of about $15.83/6.33 \approx 2.5$, whereas a non-event contributes a factor only infinitesimally below one. Thus evidence accrues in exactly the way a safety analyst would hope: rare but informative adverse events cause visible jumps, while the vast majority of non-events gradually erode over-optimistic alternatives. The e-process records the balance of those two forces over time.

This also explains why a portfolio is sensible. Different AESI have different baseline frequencies and different effect sizes. CVST is rarer but more enriched; pulmonary embolism is more common but less enriched; GBS lies somewhere in between. Expressing every stream as a valid one-step e-factor puts them on a common evidence scale. A portfolio rule can then reallocate inferential capital across streams without needing them to share the same incidence or the same effect size.

C. Why a portfolio of e-values is still valid

The next question is how to combine many streams without sacrificing validity. Let

$$\mathcal{W} = \{w \in [0, 1]^m : \sum_{j=1}^m w_j \leq 1\} \quad (8)$$

be the admissible set of portfolio weights. The leftover mass $1 - \sum_j w_j$ is interpreted as “cash” held in the neutral asset 1. If at time $t - 1$ we choose a predictable weight vector $w_{t-1} \in \mathcal{W}$, define the one-step portfolio e-factor by

$$M_t(w_{t-1}) := 1 - \sum_{j=1}^m w_{t-1,j} + \sum_{j=1}^m w_{t-1,j} E_{t,j}. \quad (9)$$

The corresponding global e-process is

$$K_t^\pi := \prod_{s=1}^t M_s(w_{s-1}), \quad K_0^\pi = 1. \quad (10)$$

Proposition 1 (Predictable portfolios remain e-processes). *If each $E_{t,j}$ is a one-step e-factor and $w_{t-1} \in \mathcal{W}$ is $\mathcal{F}_{t-1}$ -measurable, then $(K_t^\pi)_{t \geq 0}$ is a non-negative supermartingale under the global null. Consequently,*

$$\mathbb{P}_0 \left(\sup_{t \geq 0} K_t^\pi \geq \frac{1}{\alpha} \right) \leq \alpha. \quad (11)$$

Proof. By linearity of conditional expectation,

$$\mathbb{E}_0[M_t(w_{t-1}) \mid \mathcal{F}_{t-1}] = 1 - \sum_j w_{t-1,j} + \sum_j w_{t-1,j} \mathbb{E}_0[E_{t,j} \mid \mathcal{F}_{t-1}] \quad (12)$$

$$\leq 1. \quad (13)$$

Therefore,

$$\mathbb{E}_0[K_t^\pi \mid \mathcal{F}_{t-1}] = K_{t-1}^\pi \mathbb{E}_0[M_t(w_{t-1}) \mid \mathcal{F}_{t-1}] \leq K_{t-1}^\pi, \quad (14)$$

so (K_t^π) is a non-negative supermartingale. Ville’s inequality then gives the boundary-crossing guarantee. $\square$

Proposition 1 captures the conceptual separation on which the rest of the paper rests. *Validity* comes from the supermartingale structure and is unaffected by how the weights are learned, provided the rule is predictable. *Efficiency* is a separate design question: how should the weights be chosen so that a real signal is found quickly? This separation is particularly attractive in vaccine safety. Analysts may use a simple shrinkage rule, a Bayesian model, or a machine-learned predictor to decide where to place inferential capital, while the validity guarantee remains unchanged.

III. LEARNING TO STOP: THE OPTIMIZATION OBJECTIVE

A. Minimizing expected stopping time

The natural operational target is a hitting time. For a portfolio rule $\pi = (w_0, w_1, \dots)$, define

$$\tau_\alpha(\pi) := \inf \left\{ t \geq 1 : K_t^\pi \geq \frac{1}{\alpha} \right\}. \quad (15)$$

The surveillance problem is then

$$\min_{\pi \in \Pi} \mathbb{E}_1[\tau_\alpha(\pi)], \quad (16)$$

where Π is the set of admissible predictable portfolio rules and $\mathbb{P}_1$ denotes a working alternative under which one or more safety streams are elevated.

Writing the problem this way is important conceptually. In safety surveillance, the system is not rewarded for generating high expected evidence at some arbitrary fixed horizon if that evidence arrives too late to matter. The object that matters is time to alarm. The optimization criterion in (16) therefore matches the public-health question directly.

B. From hitting times to Kelly-optimal log growth

The connection to Kelly’s criterion follows from a standard large-threshold argument. Let $G_t = \log K_t^\pi$. Then

$$\tau_\alpha(\pi) = \inf \{ t \geq 1 : G_t \geq \log(1/\alpha) \}. \quad (17)$$

If the portfolio rule is stationary enough that G_t/t stabilizes at a positive growth rate $g(\pi)$ under $\mathbb{P}_1$, then for large thresholds,

$$\tau_\alpha(\pi) \approx \frac{\log(1/\alpha)}{g(\pi)}, \quad g(\pi) = \lim_{t \rightarrow \infty} \frac{1}{t} \mathbb{E}_1[G_t]. \quad (18)$$

Thus minimizing expected time to threshold crossing is approximately equivalent to maximizing long-run expected log growth:

$$\max_{\pi \in \Pi} \mathbb{E}_1[\log M_t(w_{t-1})]. \quad (19)$$

This is the Kelly criterion in portfolio language [8], [9]. The classical financial interpretation is that a growth-optimal bettor reaches large wealth targets as quickly as possible in expectation; here the same logic says that a growth-optimal e-value portfolio reaches the alarm boundary as quickly as possible in expectation.

The analogy is not merely rhetorical. If we define the one-step excess returns $R_t = (E_{t,1} - 1, \dots, E_{t,m} - 1)^\top$, then (9) becomes

$$M_t(w_{t-1}) = 1 + w_{t-1}^\top R_t. \quad (20)$$

For small returns,

$$\log(1 + x) \approx x - \frac{1}{2}x^2, \quad (21)$$

and therefore, conditionally on the past,

$$\mathbb{E}_1[\log M_t(w_{t-1}) \mid \mathcal{F}_{t-1}] \approx w_{t-1}^\top m_{t-1} - \frac{1}{2} w_{t-1}^\top A_{t-1} w_{t-1}, \quad (22)$$

where

$$m_{t-1} = \mathbb{E}_1[R_t \mid \mathcal{F}_{t-1}], \quad A_{t-1} = \mathbb{E}_1[R_t R_t^\top \mid \mathcal{F}_{t-1}]. \quad (23)$$

This produces the approximate one-step optimizer

$$w_{t-1}^* \in \arg \max_{w \in \mathcal{W}} \left\{ w^\top m_{t-1} - \frac{1}{2} w^\top A_{t-1} w \right\}. \quad (24)$$

In practice m_{t-1} and A_{t-1} are unknown and must be estimated. The important point is that the surveillance problem admits a clean optimization formulation: *learn the portfolio that makes evidence grow fastest, subject to preserving validity*.

This also clarifies why the setting differs from a bandit problem. In our case all streams are observed after each vaccinated individual, regardless of which weights were chosen. There is therefore no exploration penalty created by the act of monitoring itself. The portfolio rule decides how evidence is combined, not which data become observable. That makes a myopic log-growth objective much more defensible than it would be in settings where the action changes the information structure.

IV. ASTRAZENECA AS AN EXPOSURE-TO-ALARM CASE STUDY

A. Historical motivation and timeline

The Oxford–AstraZeneca episode offers an unusually informative test bed for multi-stream surveillance. The case is well known, but the timeline is worth restating. The vaccine was authorized by the EMA on 29 January 2021 [1]. Administration in the EU/EEA began in the first week of February 2021. On 11 March 2021, amid accumulating case reports, Denmark announced a temporary suspension and the EMA stated that 30 thromboembolic events had been reported among close to 5 million vaccinated people in the EEA [2]. Over the following days other countries, including Germany, France, Italy and Spain, paused the vaccine. The EMA’s Pharmacovigilance Risk Assessment Committee (PRAC) subsequently discussed a search on 11 March 2021 that found 269 thromboembolic cases overall, and a signal-assessment report referred to more than 5.5 million administered doses in the EU/EEA by that point [11]. On 18 March 2021 the EMA concluded that the vaccine’s benefits continued to outweigh its risks, while acknowledging a possible association with rare clotting events [12].

TABLE I: Historical timeline for the Oxford–AstraZeneca case in Europe.

Date	Event
29 Jan 2021	EMA grants conditional marketing authorization for the vaccine.
First week of Feb. 2021	Rollout begins in the EU/EEA.
11 Mar 2021	Denmark pauses the vaccine; EMA states that 30 thromboembolic events have been reported among close to 5 million vaccinated people in the EEA.
15 Mar 2021	Germany, France, Italy, Spain and others announce temporary suspensions.
18 Mar 2021	EMA concludes that benefits still outweigh risks while continuing review of rare clotting events with thrombocytopenia.

This paper does not attempt to reconstruct the committee process or the exact calendar day on which a counterfactual regulator would have acted. Instead we ask a more focused question: *how many exposures would have accrued before a valid joint monitor crossed its safety threshold?* For that estimand, the AstraZeneca episode is ideal because a later GVDN analysis provides a coherent 13-stream observed-versus-expected table using the same AESI taxonomy that is relevant to routine surveillance [3], [4].

B. Study design and assumptions

Our retrospective study design is an exposure-to-alarm simulation. We simulate vaccinated individuals one by one. For each individual and each AESI stream j , we draw a Bernoulli event indicator using a null probability p_{0j} or an alternative probability p_{1j} . The null probabilities come from the expected counts in the GVDN study; the alternative probabilities come from the corresponding observed counts. If E_j and O_j denote those counts and $N = 12,000,000$ is the denominator used to calibrate the study rates, then

$$p_{0j} = \frac{E_j}{N}, \quad p_{1j} = \frac{O_j}{N}. \quad (25)$$

The 12-million denominator is used only for *rate calibration*. It is not the historical benchmark in the headline comparison. The benchmark is the roughly 5–5.5 million exposures that had accrued before the first major European pauses.

The simplifying assumptions are deliberate. Events are treated as approximately independent across individuals and across streams. For rare complications this is a reasonable first approximation, and it matches the estimand of interest. We are not claiming that thrombotic and neurological events literally occurred instantaneously at vaccination, nor that all operational frictions can be ignored. Rather, we first estimate the number of exposures needed for a purely statistical threshold crossing and then separately apply a conservative lag correction to make contact with the historical record. Importantly, the historical pause benchmark also includes people who had

only recently been vaccinated and whose complications, if any, would not yet have declared. Thus a lag-adjusted exposure count is the right object for comparison.

C. Streams and rates

Table II summarizes the 13 AESI streams used in the simulation. Several are effectively neutral in the observed-versus-expected study; several are modestly elevated; and a few are sharply elevated. That heterogeneity is exactly what makes a portfolio approach attractive.

D. Monitoring rules

We compare several rules.

- 1) **Learned portfolio.** A data-adaptive portfolio that estimates elevated-risk probabilities $q_{t-1,j}$ stream by stream using a one-sided Beta-Bernoulli update,

$$q_{t-1,j} = \max \left\{ p_{0j}, \frac{a_j + S_{t-1,j}}{a_j + b_j + t - 1} \right\}, \quad (26)$$

where $S_{t-1,j} = \sum_{s=1}^{t-1} Y_{s,j}$. The portfolio weights are then chosen using the approximate log-growth criterion in (24) over $\mathcal{W}$.

- 2) **Uniform portfolio.** A fixed, equal-weight e-value portfolio that spreads inferential capital uniformly across the 13 streams.
- 3) **Single-stream monitors.** Standalone e-processes for each AESI, reported to show which individual channels are most informative.

In all cases the alarm boundary is $1/\alpha = 20$, corresponding to $\alpha = 0.05$.

The learned portfolio is not “adaptive-adaptive” in any opaque sense. It is simply a monitor that learns where the signal appears to be and reallocates inferential capital accordingly. Because all streams are observed every period, this learning is computationally easy and statistically legitimate.

E. Conservative lag adjustment

The immediate-observation simulation answers a statistical question: after how many exposures would the e-process cross if outcomes were observed immediately? To compare with real operations, we apply a conservative lag correction. Among the elevated serious streams we monitor, GBS has one of the longest reported onset delays after ChAdOx1 vaccination, with summaries placing the median or mean onset at roughly 11.5–13.2 days [13], [14]. By contrast, idiopathic thrombocytopenia is typically reported sooner, pulmonary embolism is around 11 days, and CVST/VITT is often reported within roughly 8–12 days [3], [12]. Using GBS as the anchor therefore makes the adjustment deliberately conservative.

Let N_{imm} be the immediate alarm count, L the conservative onset lag, and λ the early-rollout vaccination rate. Using the EMA-referenced figure of more than 5.5 million

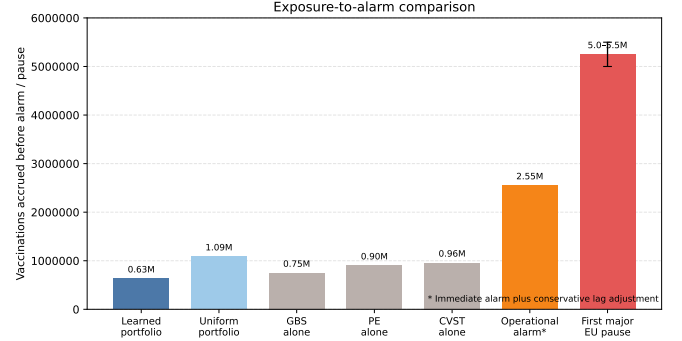


Fig. 1: Exposure-to-alarm comparison. The learned e-value portfolio reaches a valid statistical alarm after about 0.63 million vaccinations under immediate observation. Adding a conservative lag adjustment anchored to GBS onset gives an operational alarm at about 2.55 million, rounded in the text to 2.6 million. The first major European pauses occurred after roughly 5–5.5 million exposures had accrued.

EU/EEA doses by 11 March 2021 and the first week of February as rollout start implies roughly

$$\lambda \approx 1.4 \times 10^5 \text{ vaccinations per day.} \quad (27)$$

We therefore define the operational alarm count by

$$N_{\text{oper}} \approx N_{\text{imm}} + \lambda L. \quad (28)$$

With $N_{\text{imm}} \approx 0.63$ million and L at the conservative upper end of the GBS range, this yields an operational figure of about 2.5–2.6 million. Throughout the paper we report the rounded headline value *about 2.6 million*.

V. RESULTS

A. Immediate and operational alarm counts

The learned portfolio reaches the alarm threshold first. In the immediate-observation simulation it crosses at approximately 631,851 vaccinated individuals. The uniform portfolio crosses later, at approximately 1,085,794. Among the serious standalone channels, GBS is the strongest early signal, crossing at approximately 747,518; pulmonary embolism crosses at approximately 900,160; and CVST crosses at approximately 957,504. Figure 1 summarizes the main comparison.

The operational comparison is the headline claim. Once the conservative lag adjustment is applied, the learned portfolio would still have crossed after about 2.6 million exposures. The historical benchmark is 5–5.5 million. In other words, even after slowing the counterfactual alarm to account for delayed symptom onset, the learned portfolio still signals at roughly half the exposure count that had accrued by the time of the first major European pause.

B. Why the portfolio helps

The gain does not come from any one dramatic endpoint. It comes from learning how to combine several informative streams while avoiding over-commitment to the quiet ones. GBS is a powerful channel, but it is not the whole story.

TABLE II: Calibration of the 13 first-dose Oxford–AstraZeneca AESI streams. The denominator $N = 12,000,000$ is used only to convert the GVDN observed and expected counts into individual-level probabilities for the simulation.

AESI stream	Expected count	Observed count	O/E	Null rate / 1M	Observed rate / 1M
Guillain–Barré syndrome	76	190	2.50	6.33	15.83
Transverse myelitis	8	15	1.88	0.67	1.25
Bell’s palsy	260	260	1.00	21.67	21.67
Acute disseminated encephalomyelitis	6	13	2.17	0.50	1.08
Febrile seizures	15	15	1.00	1.25	1.25
Generalized seizures	3000	3000	1.00	250.00	250.00
Thrombocytopenia	1800	1926	1.07	150.00	160.50
Idiopathic thrombocytopenia	220	308	1.40	18.33	25.67
Pulmonary embolism	2600	3120	1.20	216.67	260.00
Cerebral venous sinus thrombosis	21	69	3.29	1.75	5.75
Splanchnic vein thrombosis	300	300	1.00	25.00	25.00
Myocarditis	50	68	1.36	4.17	5.67
Pericarditis	250	323	1.29	20.83	26.92

TABLE III: Key exposure counts in the retrospective replay.

Monitor or benchmark	Exposures
Learned portfolio (immediate)	631,851
Uniform portfolio (immediate)	1,085,794
GBS alone	747,518
Pulmonary embolism alone	900,160
CVST alone	957,504
Learned portfolio (operational)	≈ 2.6 million
First major EU pause	≈ 5 – 5.5 million

CVST is rarer but highly enriched. Pulmonary embolism and idiopathic thrombocytopenia contribute additional mass. A uniform portfolio dilutes these signals by allocating equal inferential capital to streams that are essentially neutral. A learned portfolio, by contrast, can concentrate progressively on the streams whose one-step e-factors show repeated deviations from background. Because validity is guaranteed by Proposition 1, this reallocation improves efficiency without endangering false-alarm control.

The result is best read as a statement about the *design of surveillance*. Endpoint-by-endpoint monitoring treats the evidence streams as separate silos. A portfolio monitor treats them as parts of one decision problem. In settings where no single harm dominates from the outset, the latter architecture should be expected to win.

C. Interpretation of the lag-adjusted comparison

The lag adjustment deserves careful interpretation. We are not asserting that a regulator would have acted the instant the statistical threshold crossed, nor that all real-world delays are captured by a single onset parameter. Rather, we are making a conservative like-for-like exposure comparison. The historical 5–5.5 million benchmark already includes individuals vaccinated only days before the pause. Their possible complications had not all been observed yet. The relevant comparison is therefore not between an immediate statistical crossing and a delayed operational pause, but between a lag-adjusted exposure count for the monitor and the exposure count actually accrued by the time the pause occurred. On that scale the learned portfolio remains materially earlier.

VI. SENSITIVITY, ROBUSTNESS, AND PRACTICAL DEPLOYMENT

A. How sensitive is the headline comparison to the lag?

The conservative GBS-based lag is intentionally chosen to avoid overstating the gain. It is helpful to understand the arithmetic. At an early-rollout pace of roughly 1.4×10^5 vaccinations per day, each additional day of lag adds about 0.14 million exposures to the operational alarm count. Hence a 10-day lag would add about 1.4 million exposures, whereas a 13-day lag adds about 1.8 million. Starting from the immediate alarm at 0.63 million, this places the operational alarm in the rough range of 2.0–2.5 million for most plausible elevated-stream lags, with our reported 2.6 million figure reflecting a deliberately conservative upper-end adjustment. The main qualitative conclusion therefore does not hinge on a knife-edge choice of lag.

This robustness matters because symptom-onset distributions are themselves uncertain and heterogeneous. Some severe outcomes may be recognized sooner, others later, and formal case confirmation can take longer still. But the gap to the real-world benchmark is large enough that moderate perturbations of the lag do not alter the substantive message. Even a fairly conservative correction leaves the portfolio alarm well below the 5–5.5 million exposures accrued before the first major European pause.

B. Dependence, background drift, and model misspecification

The simulation assumes independent Bernoulli outcomes across individuals and streams. Real data are more complicated. Individuals can experience correlated diagnoses, background rates can vary by age, sex, geography, season, and pandemic phase, and operational databases are not measured without error. None of these facts invalidates the e-value framework itself. They simply affect how the stream-specific e-factors should be built. In a deployment setting one would replace the simple Bernoulli model with self-controlled, regression-adjusted, or stratified e-factors that remain valid under the desired null. The portfolio layer would then sit on top unchanged.

This is one of the practical strengths of the approach. A portfolio e-process is agnostic about the origin of its component e-values. Those components may come from simple observed-versus-expected calculations, from stratified Poisson models, from self-controlled case series, or from cross-fitted machine-learning predictors. As long as each stream contributes a valid one-step e-factor, the combined process remains valid. This modularity is valuable in pharmacovigilance, where data pipelines and preferred epidemiologic designs differ across countries and institutions.

C. Thresholds, severity, and decision rules

We used the conventional boundary $1/lpha = 20$ with $lpha = 0.05$, but the framework is not tied to that choice. A more conservative agency could raise the threshold; a system designed for rapid triage could lower it. Raising the boundary increases absolute time to alarm for every rule, yet the relative advantage of the learned portfolio is governed primarily by its higher log-growth rate. The same logic also allows more elaborate objectives. A regulator may wish to overweight catastrophic harms relative to milder harms, or to reserve more “cash” during periods of high background instability. Such preferences can be encoded directly in the portfolio constraints or in the utility used to learn the weights.

Importantly, the statistical alarm should not be conflated with an automatic policy halt. In practice it is better understood as a pre-specified trigger for intensified review: a signal that the joint evidence across multiple harms has become too strong to ignore. That is exactly how many agencies already behave informally. The value added here is a formal, anytime-valid rule for when such escalation becomes warranted.

D. What deployment would look like

A deployable system would be simple to describe. First, specify the AESI panel, background-rate model, risk windows, and stream-specific e-factors. Second, choose the alarm threshold and the admissible portfolio class. Third, as new vaccination records and linked outcomes arrive, update each stream’s one-step e-factor and then update the global e-process using the current predictable weights. Fourth, display both the global evidence trajectory and the channel-level contributions. Because the portfolio uses all streams simultaneously, the dashboard can show not only whether the system is close to alarm, but also which harms are driving growth.

Such a system could be run daily or weekly, centrally or in federated fashion. It would also be auditable. Every threshold crossing could be traced back to a sequence of publicly documented weight updates and channel-level e-factors. This is attractive institutionally: the same mechanism that improves statistical efficiency also improves transparency.

VII. RELATION TO EXISTING SURVEILLANCE PARADIGMS

The proposed framework is best viewed as a new layer on top of, rather than a replacement for, standard

pharmacovigilance practice. Classical post-marketing safety monitoring often relies on disproportionality analyses in spontaneous reporting systems, observed-versus-expected comparisons in registries, or sequential score and likelihood-ratio monitoring such as CUSUM and maxSPRT [?]. These tools have been highly influential, but they often treat outcomes one at a time or require a fixed testing plan that sits uneasily with modern adaptive workflows.

E-values differ in two ways. First, they are inherently anytime-valid, so the same evidence object remains interpretable under optional continuation, repeated looks, or online adaptation. Second, they compose naturally. Stream-specific e-factors can be multiplied across time and pooled across outcomes without leaving the validity framework. This composability is what makes a portfolio view possible. In contrast, combining many sequential likelihood-ratio or p-value streams usually requires either explicit multiplicity correction or a more delicate alpha-spending architecture.

The paper’s contribution is therefore not another single-endpoint sequential test. It is a method for *joint* surveillance when signals may be diffuse, heterogeneous, and only partially aligned across endpoints. In this sense the work is closer in spirit to the literature on online multiple testing and e-value aggregation than to classical one-stream surveillance alone [?], [7]. At the same time, the optimization layer links it to the growth-optimal portfolio literature. The resulting combination—validity from e-processes and efficiency from Kelly-style log-growth maximization—appears to be particularly well suited to post-deployment vaccine safety.

The approach should also be compared with full Bayesian decision systems. Bayesian monitoring can be extremely powerful, especially when prior structure across related outcomes is strong. But unless additional calibration steps are taken, posterior probabilities do not automatically deliver frequentist false-alarm guarantees under optional stopping. E-value portfolios offer a complementary route: they keep a clear pathwise guarantee and can still absorb Bayesian ideas through the choice of working alternatives, shrinkage rules, or predictive models used to construct the stream-specific e-factors.

VIII. DISCUSSION

The AstraZeneca case study suggests a broader methodological point. Modern safety surveillance often inhabits an uncomfortable space between two statistical cultures. One culture prizes flexibility and real-time judgment but risks losing inferential calibration when analyses are repeated or adapted. The other culture prizes formal guarantees but is often designed around a single endpoint and a fixed horizon. E-value portfolios offer a way to keep the first culture’s flexibility while recovering the second culture’s honesty.

Several features make the approach especially appealing for pharmacovigilance. First, the evidence object is *anytime-valid*. Committees may look repeatedly, analysts may update models online, and a formal alarm can still be interpreted at face value because the false-alarm probability

is controlled pathwise by Ville’s inequality. Second, the framework is naturally *modular*. The validity layer depends only on stream-specific e-factors and predictable weighting. The efficiency layer can be improved over time without changing the core guarantee. Third, the optimization objective is operationally meaningful. By formulating surveillance as minimizing expected stopping time, the paper makes clear why log-growth ideas from Kelly-optimal investing belong in this domain. The connection is not metaphorical; it is the right mathematical approximation to the alarm-timing problem.

The study design also deserves comment. We intentionally frame the analysis as an exposure-to-alarm simulation rather than apologize for not reproducing calendar time exactly. A full historical replay would require day-by-day vaccination totals, reporting delays, medical attention delays, case adjudication rules, data-linkage lags, and an explicit model of how committees act on statistical evidence. Such a reconstruction is interesting, but it is a different project. The question studied here is narrower and cleaner: if the post-vaccination world had looked like the observed-vs-expected rates in the GVDN study, how many vaccinations would have accrued before a valid joint monitor raised a flag? For that question, the stylized design is not a flaw; it is an appropriate abstraction.

This does not mean the exercise is assumption-free. The simulation treats event indicators as independent Bernoulli draws and does not model within-person clustering or changing background incidence over calendar time. The design probabilities are learned through a simple one-sided Beta-Bernoulli mechanism rather than a richer hierarchical model. We also leave aside estimation uncertainty in the background rates and in the observed-vs-expected table, because the goal of the paper is to isolate the portfolio idea itself. These are meaningful limitations, but none of them changes the qualitative message.

That message is also broader than the AstraZeneca episode. Vaccines, post-marketing drug surveillance, medical-device monitoring, hospital quality surveillance, and even cyber-security alerting all share the same structural problem: multiple weak streams, optional looks, and high value placed on early but trustworthy action. E-value portfolios are a general-purpose answer to that problem. The specific learned rule used here is only a first implementation. One could imagine richer Bayesian shrinkage across clinically related AESI, severity-weighted objectives, covariate-adjusted stream models, or hybrid systems that incorporate spontaneous reports and registry data simultaneously.

Finally, there is a civic reason to care. Rare safety signals are difficult not only statistically but socially. When a concern emerges, authorities are under simultaneous pressure not to overreact and not to react too late. A method that separates validity from efficiency is valuable precisely because it lowers that tension. One can search aggressively for early evidence while keeping a clean, public-facing false-alarm guarantee. In that sense, learning to stop is not merely a technical problem. It is part of how

institutions learn to act credibly under uncertainty.

IX. CONCLUSION

We presented a portfolio view of sequential vaccine-safety monitoring based on e-values. The framework treats each adverse-event stream as a valid source of sequential evidence, combines those streams through predictable portfolio weights, and chooses the weights to approximately minimize time to alarm by maximizing log growth. The resulting monitor remains anytime-valid under continuous monitoring and optional stopping.

In an AstraZeneca-calibrated exposure-to-alarm simulation, the learned portfolio crosses its statistical boundary after about 0.63 million vaccinations under immediate observation and after about 2.6 million vaccinations under a conservative lag adjustment anchored to GBS onset. This compares with roughly 5–5.5 million exposures before the first major European pause in March 2021. The substantive conclusion is therefore clear: better evidence allocation across multiple harms could, in principle, have made the signal visible materially earlier. The methodological conclusion is equally clear: in sequential safety surveillance, learning to stop is naturally a Kelly-optimal e-value portfolio problem.

APPENDIX A

WHY MINIMIZING TIME TO THRESHOLD IMPLIES MAXIMIZING LOG GROWTH

For completeness, we sketch the approximation behind (18). Write $G_t = \log K_t^\pi = \sum_{s=1}^t \log M_s(w_{s-1})$. If, under a stationary or slowly varying alternative, $t^{-1}G_t \rightarrow g(\pi) > 0$ in probability and in mean, then for large boundaries the crossing time of the level $b = \log(1/\alpha)$ satisfies $\tau_\alpha(\pi) \approx b/g(\pi)$. Intuitively, the path of G_t becomes approximately linear with slope $g(\pi)$, so the time needed to reach height b is approximately $b/g(\pi)$. Since b does not depend on the portfolio rule, minimizing the hitting time is approximately the same as maximizing the asymptotic slope. This is the same argument that underlies the classical connection between Kelly-optimal investing and fastest growth to large wealth targets [8], [9]. In the present application the target is not a wealth level but an evidence threshold.

APPENDIX B

WHY “THE FIRST STREAM TO CROSS” IS NOT THE SAME AS A VALID GLOBAL ALARM

It is tempting to monitor each AESI separately and raise a global alarm when the first one crosses $1/\alpha$. That procedure is generally not level- α for the global null, because it creates a multiplicity problem across streams. A conservative repair is to give each of the m streams its own threshold m/α , so that a union bound plus Ville’s inequality controls the probability that any stream ever crosses. But that repair is conservative and wastes the possibility of pooling weak evidence across endpoints. The portfolio construction resolves the problem in a more efficient way: it pools e-values before thresholding, producing a single global

e-process whose validity does not require a Bonferroni penalty.

APPENDIX C

A MINIMAL IMPLEMENTATION RECIPE

A practical implementation can be summarized in five steps.

- 1) Choose the AESI panel, risk windows, and null models p_{0j} or their more elaborate time-varying analogues.
- 2) For each stream, define a valid one-step e-factor $E_{t,j}$ against upward risk shifts.
- 3) Specify the admissible portfolio class $\mathcal{W}$ and a predictable learning rule for w_{t-1} , such as the quadratic log-growth optimizer in (24).
- 4) Update the global e-process recursively as $K_t^\pi = K_{t-1}^\pi M_t(w_{t-1})$ and trigger a formal alarm when $K_t^\pi \geq 1/\alpha$.
- 5) After an alarm, inspect the channel-level contributions, evaluate data quality and case adjudication, and decide whether escalation, communication, or temporary restriction is warranted.

The recipe is intentionally generic. Its role is not to prescribe one unique workflow, but to show that the gap between statistical theory and operational pharmacovigilance is narrower than it may appear.

APPENDIX D

WORKED ILLUSTRATION: EVIDENCE ACCUMULATION IN THE GBS STREAM

A small worked example helps make the e-value mechanics concrete. For GBS, the calibration in Table II gives a null risk of $p_0 \approx 6.33 \times 10^{-6}$ per vaccinated individual and a working alternative near $q \approx 15.83 \times 10^{-6}$. If a GBS case occurs, the one-step factor in (5) contributes approximately $q/p_0 \approx 2.5$. If no case occurs, the factor contributes approximately $(1 - q)/(1 - p_0) \approx 0.9999905$. Thus one rare case generates a visible jump, while millions of non-cases produce only a slow drift.

Suppose for simplicity that the design alternative stays fixed at q . After n vaccinated individuals and x observed GBS cases, the GBS e-process is

$$K_n^{\text{GBS}} = \left(\frac{q}{p_0}\right)^x \left(\frac{1 - q}{1 - p_0}\right)^{n-x}. \quad (29)$$

Taking logs shows the two ingredients clearly:

$$\log K_n^{\text{GBS}} = x \log \frac{q}{p_0} + (n - x) \log \frac{1 - q}{1 - p_0}. \quad (30)$$

The first term rewards enriched events; the second term penalizes alternatives that expect too many events. This is why single-stream evidence can remain modest for a long time even when the rate ratio is substantial: rare events are informative, but they are still rare.

The portfolio gains efficiency by allowing several such modestly informative streams to contribute together. A few excess GBS events, a few excess CVST events, and a modest but repeated elevation in pulmonary embolism

can jointly drive the global e-process upward even if none of the streams dominates immediately. This is the central intuition behind the paper’s main empirical finding. The learned portfolio is not discovering a hidden miracle endpoint. It is learning how to turn several weak but coherent pieces of evidence into one stronger sequential signal.

APPENDIX E

EXPOSURE-TO-ALARM VERSUS CALENDAR TIME

Our estimand is the number of vaccinations accrued before an alarm, not the exact date of a regulatory decision. The distinction matters. A full calendar-time replay would require daily vaccination counts, symptom-onset distributions, reporting delays, medical attention delays, adjudication rules, and a model of how regulators transform statistical evidence into action. Such a replay is feasible in principle but would obscure the main methodological point of the paper, namely how to combine multiple evidence streams while preserving anytime-validity.

The exposure-to-alarm estimand has three practical advantages. First, it is directly interpretable in public-health terms: fewer exposures before alarm means fewer people put at risk before a signal becomes visible. Second, it is comparable across countries and deployment schedules, whereas the same number of days can correspond to very different exposure counts when rollout speed differs. Third, it matches the way the historical benchmark is usually reported. The mid-March 2021 AstraZeneca discussions were framed in terms of the number of administered doses or vaccinated individuals—roughly 5 to 5.5 million in the EU/EEA—rather than in terms of a common day count from launch. For those reasons, exposure-to-alarm is not merely a convenient proxy; it is arguably the more relevant quantity for retrospective method evaluation.

APPENDIX F

OPERATIONAL ARITHMETIC OF THE LAG ADJUSTMENT

Equation (28) makes transparent how the lag enters. If L is measured in days and λ denotes vaccinations per day during the early rollout, then changing the lag by one day changes the operational alarm by approximately λ . With $\lambda \approx 1.4 \times 10^5$, even a three-day difference between two plausible serious-stream lags changes the operational count by only about 0.42 million exposures. This is useful for interpretation. The exact onset distribution matters, but the order-of-magnitude comparison does not rest on any implausibly precise lag estimate. The conservative GBS-anchored adjustment is therefore best seen as a transparent, easy-to-audit sensitivity choice rather than a hidden modelling assumption.

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